Study Guide

COMPOSITION OF EXAM

Lecture topic	Points	# of Qs	% Total points	% Total questions
Internal Regulation	40	14	***29.63***	***32.56***
Learning & Memory	33	10	***24.44***	***23.26***
Emotional Behaviors	36	11	***26.67***	***25.58***
Psychological Disorders	26	8	***19.26***	***18.60***
TOTAL	**135**	**43**	**100.00**	**100.00**

Homeostasis/Internal Regulation

Homeostasis and allostasis

- Define homeostasis and how it is achieved (effectors, sensors, negative feedback).
  - tendency of an organism to regulate its internal environment and maintain equilibrium (receptor -> integrator -> effector)
- Be able to distinguish the difference between homeostasis and allostasis
  - Allostasis is **an adaptive** process in which the body prepares in a **predictive** way to an **anticipated need** or challenge
  -

Temperature regulation

- Regulated by the POA/AH
- receptor -> integrator -> effector
  - Think thermostat
- Receptors located in skin, spinal cord, brain

Body temp above "set point"	**Body temp below "set point"**
Blood vessels near skin dilate	Blood vessels near skin constrict
Sweat glands stimulated	Skeletal muscles cause shivering
Respiratory centers stimulated to cause panting	Chemical signals increase cellular respiration & heat production

- Fever
  - Way body fights infection
  - Know what the role is of the hypothalamus (AH/POA) in fever.

Thirst

Type of Thirst	Osmotic	Hypovolemic (thirst based on low volume)
Causes	high solute concentration outside cells causes ”Cell dehydration”	low blood volume
Best relived by drinking	water	water containing solutes (e.g. salt water or sport drinks)
Receptor location	**osmotic receptors** in OVLT & SFO, brain areas adjoining the third ventricle	**baroreceptors** in blood vessels, heart, kidney
Hormone influences	**vasopressin (ADH) secretion** to conserve water via kidneys	**angiotensin II secretion** to maintain sodium and blood levels

- If you lack vasopressin, you would drink more pure water

Hunger

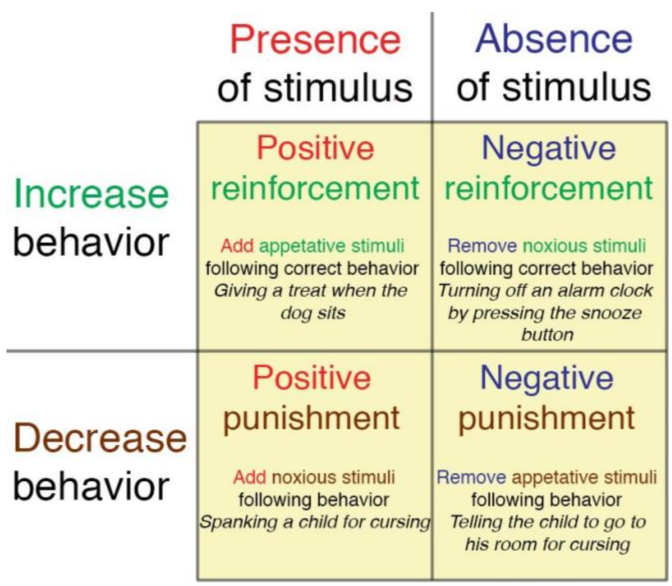
- Digestion
    - What is the sequence of events when we digest food?
    - Vagus nerve conveys information about satiety to the hypothalamus via sensing of stomach distention that involves CCK release from the duiodenum and then the Vagus nerve causes the hy[pthal;amus to release a shorter form of CCK.
- Insulin and glucagon (Secreted by pancreas)
    - Insulin - enables glucose to enter the cells (pre and during meal)
    - Glucagon- stimulates the liver to convert some of its stored glycogen back to glucose (post-meal).
    - Diabetes – type 1: low levels of insulin, with high levels of glucose and hunger
- Appetite network
    - Know the different peripheral signals that send information to the hypothalamus and what type of information they send.
        - Leptin –signals about fat reserves; defects -> overeating; secreted by fat cells;

        - Ghrelin stimulates appetite; secreted by stomach
        - PYY – suppresses appetite; secreted by intestines
    - Know what happens to hunger when you damage or activate the lateral hypothalamus.
        - Lateral hypothalamus: **stimulation** increases appetite; **damage** -> undereating; low insulin
- Eating disorders
    - Be able to name and describe different eating disorders we talked about in class.
    - Understand the neurobiology of overeating.
    - Obesity: Syndromal – Prader Willi; Monogenic - e.g. *melanocortin receptor gene mutation*; Polygenic – common obesity
    - Bulimia – changes in brain region that are activated by drugs of abuse (e.g. nucleus accumbens)
    - Anorexia – possible behavioral treatment strategy is temperature regulation; keep body warm to reduce activity

Learning & Memory

Types of Learning and Conditioning

- Classical (Pavlovian) – relationship between 2 events
    - **Association** between neutral stimulus (*NS*) and unconditioned stimulus (*UCS*); and thus to the unconditioned response (*UCR*).
- Instrumental (operant) - relationship **between an action & outcome**
    - 4 types: 2 positive & 2 negative
        - Positive Reinforcement
        - Negative Reinforcement
        - Punishment
        - Omission training



Exercise 1

Directions: Identify the UCS, UCR, NS, CS, and CR in the following situations.

Hint: Look for the UCR first, UCS next. The NS always becomes the CS.

1. A boy is fond of sour pickles, which make his mouth water whenever he eats them. He passes an open vat of pickles in the supermarket and begins to salivate.
    - **UCS:** sour pickles
    - **UCR:** mouth water

- **NS:** vat of pickles
- **CS:** vat of pickles
- **CR:** salivate

2. A boy who is trained in karate often practices by throwing mock punches at his sister. One day, he accidentally hits her in the eye and she recoils in pain. From that day on, every time he raises his hands, his sister flinches.
    - **UCS:** punched in eye
    - **UCR:** recoils in pain
    - **NS:** raising hand
    - **CS:** raising hand
    - **CR:** flinches

What type of learning is this? ____Classical Learning__

Exercise 2

In each scenario below, identify the behavior that is being rewarded/punished and the reward/punishment.
Also, identify if it is an example of positive reinforcement, negative reinforcement, negative punishment, or positive punishment.

1. Portia praises her little brother by telling him what a great job he’s done each time he takes the trash out. Now, his little brother takes the trash out each night without being asked.

Behavior: Taking trash out
Reward or Punishment?: Reward – Praise
This is **positive reinforcement** because the behavior (taking out the trash) increases as something positive (praise) is given.

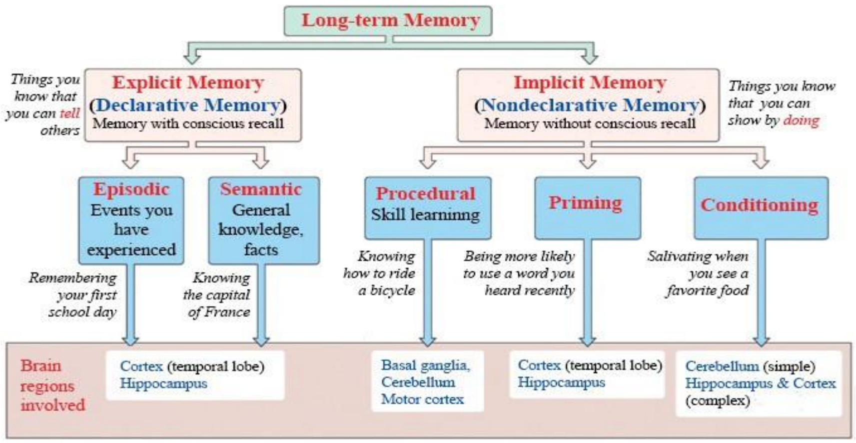
2. Xavier drives very carefully because he has gotten several traffic tickets in the past.

Behavior: Driving recklessly
Reward or Punishment?: Punishment – Traffic tickets
This is **positive punishment** because the behavior (driving recklessly) decreases as something negative (traffic tickets) is added.

3. When Kasha plays her music too loud in her room, her mom takes away her stereo system. So she tries to keep the volume down.

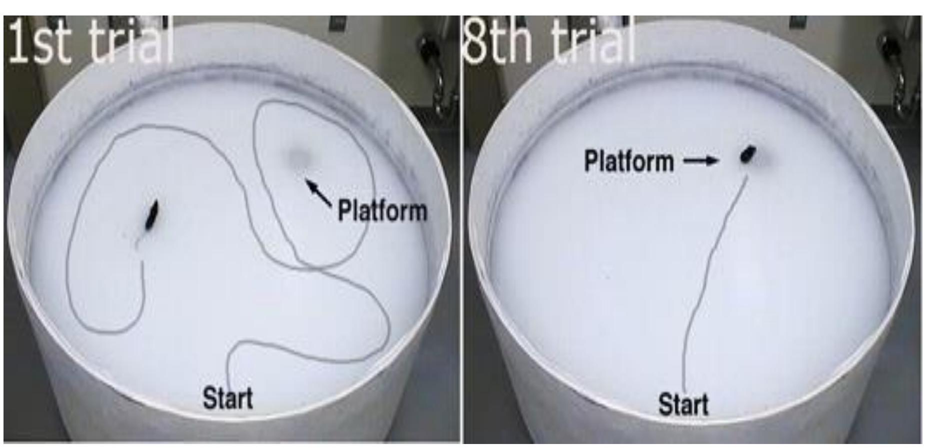
Behavior: Playing her music loudly
Reward or Punishment?: Punishment – Stereo system is taken away
This is **negative punishment** because the behavior (playing music loudly) decreases as something positive (stereo system) is being taken away.
What type of learning is this? ____Operant Learning__

TYPES OF MEMORY



Brain regions involved in learning & memory

	Hippocampus	Striatum
ROLE	formation; episodic	habit; procedural
SPEED	can be single trial	gradually with repetition
Malleability	Flexible	Rigid/automatic
Conscious recall	Yes; Explicit	No; Implicit
Effect of damage	**Impaired declarative memory**	**Impaired skill acquisition & habit formation**



Types of memory impairments

- Anterograde amnesia – inability to form new memories after damage
- Retrograde amnesia – loss of memories or previously learned information
- If you remove the hippocampi of a rodent it **will not learn** how to navigate a maze and **decrease** the time to complete a maze
    - After overtraining hippocampal damage does not decrease maze performance because it has become habitual and governed by the activity of the striatum

LTP Properties

1. Cooperativity: LTP is greater if 2 inputs are active together
2. Specificity: only stimulated synapses are affected
3. Associativity: Pairing weak input with strong input enhances response to

Emotional Behaviors

Theories of emotion

James-Lange – physiological signals generate emotions
Evidence: 1) physiological responses can influence emotions
a) Pen in mouth experiment
Issues: 1) patients w autonomic failure experience emotions (but less intense)2) similar physiological changes occur for multiple emotions
3) Unclear if there are unique universal facial expressions
4) cerebral cortex removed from cats, and still show anger

Cannon-Bard - Emotion occurs both *simultaneously* & *independently* from physiological reaction.
Evidence: Lesion studies; different brain areas involved in subjective emotions and physiological responses
Issues: doesn't account for physiological changes can alter the subjective experience of an emotion; ADRENALINE STUDY

Schacter-Singer – physiological response influences experience of emotion while accounting for context
Evidence: Adrenaline study

Physiology & emotions

- Facial expressions can affect mood
- Decreased volume and activity often seen in key brain regions affected in PTSD (except amygdala - hyperactive)
- **Botox shown to decrease depression**

Brain Mechanisms of Emotion

No brain area is responsible for **one and only one** emotion; areas associated with an emotion vary across individuals

Amygdala – necessary for fear & anxiety

- **Fear conditioning** – pair light with shock to measure how light (CS) modifies startle reflex
  - If damaged, startle reflex is same across situations (impaired associative learning)
  - Urbach-Wiethe: rare genetic condition; calcium accumulation damages amygdala -> fearlessness
  - **Different pathways of amygdala responsible for different types of fear and fear responses**

Stress responses

- Cortisol acutely increases energy & alertness
- sympathomedullary pathway (**SAM)** mediates the **fast** stress response
- **HPA** axis – **slower** reaction and last **longer**
  - Glucocorticoids are released from the adrenal gland
- Chronic stress can impair mental & physical health

Panic disorder

- Frequent periods of anxiety and occasional attacks of rapid breathing, increased heart rate, sweating, and trembling
- Possible genetic component
- Linked to hypothalamus abnormalities
- **Decreased GABA, increased orexin**

Anti-anxiety medications

- Benzodiazepines - most used drugs; e.g. diazepam (Valium), alprazolam (Xanax)
- Bind to the $GABA_A$ receptor, and facilitate the effects of GABA
- Exert their effects in the amygdala, hypothalamus, midbrain, and other areas

Psychological Disorders

Substance Use

Agonist (↑affinity/ ↑efficacy)	**Drugs that mimic or increase an effect of a neurotransmitter** • **fit the lock and open it easily (i.e. open the ion channel, activate second messengers, etc.)** • **mimic the actions of the endogenous neurotransmitter**

Antagonist (↑affinity/ ↓efficacy)	**Drugs that block or decrease an effect of a neurotransmitter** • **drug fits the lock but can not turn the key** • **drug binds to the receptor but can not activate it** • **prevent neurotransmitters from binding and activating it too**

Reward/reinforcement brain circuit

Ventral tegmental area

- Contains a large # of DA producing neurons
- Project widely throughout the brain
- DA neuronal activity affected by many psychoactive drugs

Nucleus accumbens (NAc)

- Central to reinforcing experiences of all types
    - Primary location where addictive drugs release dopamine and/or norepinephrine

- After people become addicted to a drug, the drug releases less dopamine, but cues associated with the drug release more

negative reinforcement theory
Taking drugs to counteract withdrawal symptoms or a negative state
Problems
Highly addictive drugs like nicotine and cocaine do not produce severe withdrawal symptoms.
People relapse long after withdrawal symptoms are gone.
Some drugs like tricyclic antidepressants produce withdrawal symptoms but do not promote addiction.
positive reinforcement theory
Taking drugs because they produce euphoric pleasure
Problems
Euphoric feeling often goes away.
Some addictive drugs do not produce pleasurable feelings.
The destruction that drug addiction produces far outweigh any pleasure from the drug.
incentive-sensitization theory of addiction
The MLDS becomes sensitized to the drugs and to stimuli associated with the drug.
When the MLDS is activated, it creates strong cravings for the drugs.
Classical conditioning of stimuli associated with taking drugs
Instrumental conditioning of the increased craving when the stimuli are present.
Sensitization is long-lasting far beyond withdrawal symptoms.
The **dorsal striatum** functions during the learning process

Mood & neurodevelopmental disorders

- MDD appears to have many causes (etiologies) but classuically was thought of as a disruption of the monoamine system
- Typical antidepressants take weeks to exert any effects
    - Their effects are thought to occur through the actions of BDNF & its receptor
- Ketamine is an anesthetic that now is used as a rapid antidepressant at subanesthetic does
    - It is considered a dissociative hallucinogen
    - One proposed mechanism for its effects is antagonism of NMDARs
    - This rapidly induced plasticity in the brain, especially compared to SSRIs
- Autism has a strong genetic component
    - Most mutations are de novo and thus rare
    - Drugs affecting oxytocin and serotonin signaling are showing promise as novel therapeutics